

Yajnaseni Dutta

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Universiteit Leiden – Mathematisch Instituut
Gorlaeus Gebouw, Rm. BW 2.34, Leiden, NL.

RESEARCH INTERESTS

Hyperkähler manifolds, generic vanishing theory, birational geometry, Hodge theory and Hodge module theory, derived categories

EMPLOYMENT

Assistant Professor (Universitair Docent) Sep. 2023–present

Universiteit Leiden, Mathematical Institute (tenured Nov. 2024)

Postdoc Sep. 2022–Aug. 2023

Rheinische Friedrich-Wilhelms-Universität Bonn, Complex Geometry Group of Prof. Daniel Huybrechts

Hausdorff Postdoc Sep. 2019–Aug. 2022

Rheinische Friedrich-Wilhelms-Universität Bonn, Complex Geometry Group of Prof. Daniel Huybrechts

SELECTED RESEARCH VISITS

Oct. 2023 Visitor (ten days), Universität Bonn. Host: Prof. Daniel Huybrechts

Oct. 2019–Dec. 2019 Visiting Scholar (three months), Katholieke Universiteit Leuven. Host: Prof. Nero Budur

EDUCATION

Sep. 2014–Aug. 2019 Ph.D. Mathematics at **Northwestern University, USA**. Advisor: Prof. Mihnea Popa

Aug. 2013–Aug. 2014 Ph.D. candidate Mathematics at **University of Illinois at Chicago, USA**. Advisor: Prof. Mihnea Popa

Aug. 2011–Jul. 2013 M.Sc., Mathematics at **Chennai Mathematical Institute, India**

Aug. 2008–Jul. 2011 B.Sc., Mathematics and Computer Science at **Chennai Mathematical Institute, India**

LIST OF PUBLICATIONS/PREPRINTS

9. **Relative compactified Prym and Picard fibrations associated to very good cubic fourfolds.** jt. with L. Marquand, to appear in Proc. Amer. Math. Soc. [arXiv:2502.21301](https://arxiv.org/abs/2502.21301)
8. **Twists of intermediate Jacobian fibrations.** jt. with D. Mattei, and E. Shinder, preprint available at [arXiv:2411.01953v1](https://arxiv.org/abs/2411.01953v1)
7. **Some density results for hyperkähler manifolds.** jt. with E. Izadi, L. Kamenova, and L. Marquand, to appear in Math. Res. Lett. [arXiv:2403.04868](https://arxiv.org/abs/2403.04868)
6. **On symplectic birational self-maps of projective hyperkähler manifolds of $K3^{[n]}$ -type.** jt. with D. Mattei and Y. Prieto-Montañez, International Mathematics Research Notices, vol. 5 (2024) [doi:10.1093/imrn/rnae112](https://doi.org/10.1093/imrn/rnae112)
5. **Maximal variation of curves on $K3$ surfaces** jt. with D. Huybrechts, Tunisian J. Math. vol. 4, 2022, no. 3 [doi:10.2140/tunis.2022.4.443](https://doi.org/10.2140/tunis.2022.4.443)

4. **Generic Vanishing, 1-forms, and Topology of Albanese Maps** jt. with F. Hao and Y. Liu. Math. Zeitschrift. Volume 306, no. 56, (2024) [doi:10.1007/s00209-024-03438-3](https://doi.org/10.1007/s00209-024-03438-3)
3. **Vanishing for Hodge ideals on toric varieties** Mathematische Nachrichten. vol. 293, 2020. [doi:10.1002/mana.201800363](https://doi.org/10.1002/mana.201800363)
2. **Effective generation and twisted weak positivity of direct images** jt. with T. Murayama. Algebra and Number theory. vol. 13, no. 2, 2019. [doi:10.2140/ant.2019.13.425](https://doi.org/10.2140/ant.2019.13.425)
1. **On the effective freeness of the direct images of pluricanonical bundles** Annales de l'Institut Fourier, vol. 70, no. 4, 2020. [doi:10.5802/aif.3351](https://doi.org/10.5802/aif.3351)

GRANTS, AWARDS, FELLOWSHIPS AND RECOGNITION

Sep. 2023	Starter budget for hiring a PhD student (€240,000), Universiteit Leiden
May 2019	Best Thesis in Mathematics, Northwestern University
May 2018	Gelfand Award, Northwestern University (awarded to the Ph.D. student who has contributed most to the department)
2013–2014	University Fellowship, University of Illinois at Chicago (nominated by the Maths Department among the top 15% of newly recruited graduate students, receiving a full year's stipend and tuition waiver)
2008–2011	Scholarship for Higher Education, Department of Science and Technology, Govt. of India (a component of the programme "Innovation in Science Pursuit for Inspired Research (INSPIRE)")
2007	Qualified regional mathematical olympiad, West Bengal (unique track for being part of the IMO-team, India and is held across regions throughout India)

THESIS SUPERVISION

Dec. 2023–present	PhD Thesis (co-advising with Prof. Evgeny Shinder): Céline Fietz, Universiteit Leiden
Sep. 2025–present	Master's Thesis: Matthijs Buijsen, Moduli of Sheaves on Threefolds, Universiteit Leiden
Mar. 2025–present	Master's Thesis (co-advising with Dr. E. Brakkee): Jorge Macein, Twisted Elliptic Fibrations of the Fermat quartic, Universiteit Leiden
Oct. 2022–Oct. 2023	Master's Thesis (co-advised with Prof. E. Shinder): Céline Fietz, Categorical Resolutions of A2 Singularities, Universität Bonn
Feb.–Jun. 2024	Bachelor's Thesis: Lucio Verrijzer, The Morita Equivalence and (Graded) Brauer Groups, Universiteit Leiden

TEACHING

Spring 2025	Instructor: Representation Theory, Universiteit Leiden
Fall 2024	Instructor: Advanced Algebraic Geometry: Moduli of sheaves, national master-math program, Vrije Universiteit Amsterdam
Spring 2024, 25	Instructor: Mathematical Reasoning, Leiden University College
Fall 2023, 25	Instructor: Algebraic Curves, Universiteit Leiden
Summer 2022	Teaching Assistant: Algebraic Geometry II, Universität Bonn
Winter 2022	Teaching Assistant: Algebraic Geometry I, Universität Bonn
Winter 2022	Hauptseminar Algebra: Klassische Algebraische Geometrie, Universität Bonn
Winter 2021	Lecture course: Selected Topics in AG: Moduli of Vector Bundles, Universität Bonn
Winter 2020	Graduate Seminar: Algebraic Curves and Abelian Varieties, Universität Bonn
Fall 2015–Spring 2018	Teaching Assistant: Calculus (all levels), Linear Algebra, Foundations of Higher Math, Intro to Topology, Graph Theory, Northwestern University
Fall 2014, Winter 2015	Grader: Differential Topology, Algebraic Geometry, Northwestern University

INVITED WORKSHOP VISITS

Jul. 2026	Algebraic Geometry: Moduli Spaces, Birational Geometry and Derived Aspects, Mathematisches Forschungsinstitut Oberwolfach
Apr. 2026	Complex Analysis, Mathematisches Forschungsinstitut Oberwolfach
Sep. 2023	Geometry of HyperKähler Varieties, IASM Hangzhou
Aug. 2023	Women in Mathematical Physics II, Banff, Canada (participated online)
Apr. 2023	Arbeitsgemeinschaft: Twistor D-Modules and the Decomposition Theorem, Mathematisches Forschungsinstitut Oberwolfach (Oberwolfach report)
Jul. 2022	Algebraic Geometry: Moduli Spaces, Birational Geometry and Derived Aspects, Mathematisches Forschungsinstitut Oberwolfach
Sep. 2021	Summer School: Irregular Riemann-Hilbert Correspondence, Centre Paul Langevin in Aussois

SELECTED INVITED TALKS

Conference talks

2026	Cubic Fourfolds, Gushel-Mukai Fourfolds, and Hyperkahler Manifolds, ICERM, October
	Higher Dimensional Varieties, Erdős Center, Budapest, October
	Rationality problems, Cremona groups, and motivic invariants, LMS symposium, Sheffield, July
	Hodge Theory, Ashoka University, Delhi, January
2025	Conference in Algebraic Geometry via Eurostar, Leiden, Netherlands, September
	Japanese European Symposium on Symplectic Varieties and Moduli Spaces, Netherlands, September
	HK days in Sheffield, Sheffield, February
2024	D-modules, local systems and applications, CRM Montreal, September
	GQT symposium, Nijmegen, August
	DIAMANT symposium, Utrecht, April
	GQT section: Netherlands Mathematical Congress, Lunteren, April
	Fano and Hyperkähler varieties, Genova, January
2023	Belgian–Dutch Algebraic Geometry Meeting, Leiden, October
	Geometry of HyperKähler Varieties, Hangzhou, September
	Derived categories, Hodge theory and Singularities, Sheffield, June
2022	Algebraic Geometry section: DMV-Jahrestagung, FU Berlin, September
	Workshop: Curves and K3 surfaces, Bonn, May
	D-Modules: Applications to Algebraic Geometry, Arithmetic and Mirror Symmetry, CIRM (Luminy), April

- 2021 Singularities in the Midwest Workshop, Wisconsin (Online), May
 2020 AMS Fall Eastern Sectional, "Algebraic Singularities in Arbitrary Characteristic", (Online), October
 Algebra, Algebraic Geometry and Number Theory section: DMV-Jahrestagung, Chemnitz (Online),
 September

Lecture Series

- Jan. 2026 Swiss-French Workshop in Algebraic Geometry, Charmey, Switzerland
 Mar. 2025 D-modules in Bochum – lecture series (3 hours), Ruhr-Universität Bochum
 Nov. 2018 D-modules in birational geometry – Short course (3 hours), University of Chicago

General Audience Talk

- 2026 D-modules and Singularities, Abel Prize Evening, Royal Netherlands Academy of Arts and Sciences
 (KNAW), February

Seminars (selection)

- 2024 Arithmetic and Algebraic Geometry seminar, Universiteit van Amsterdam, November
 Harvard–MIT Algebraic Geometry Seminar, September
 Algebraic Geometry Seminar, Johns Hopkins, September
 Séminaire "Arithmétique et géométrie algébrique" de Strasbourg, March
 Algebraic Geometry Seminar, Princeton University, February
 2023 Algebraic Geometry Seminar, USTC, Hefei, September
 Seminar: ESAGA group joint oberseminar, Essen, July
 2022 Mathematics Colloquium, Institute of Mathematical Sc., Chennai, August
 Algebraic Geometry Seminar, Columbia University, March
 2021 Algebraic Geometry Seminar, Stony Brook University (Online), September
 Algebraic Geometry Seminar, Princeton University (Online), July
 Online Algebraic Geometry Seminar, UC, San Diego, (Online), April
 Edinburgh Hodge seminar (online), January
 2020 Algebra and Geometry Seminar, Frankfurt (Online), December
 Zoom seminar in TIFR, Mumbai (Online), November
 Oberseminar Algebraic Geometry, Hannover (Online), July
 CMI Seminar, Chennai Mathematical Institute (online), June
 Seminar on Applications of Hodge modules to birational geometry, HU Berlin, Germany, January
 2019 Oberseminar "Algebra und Theoretische Mathematik", Chemnitz, Germany, November
 Seminar Number Theory and Algebraic Geometry, KU Leuven, Belgium, October
 Algebraic Geometry and Commutative Algebra Seminar, University of Notre Dame, May
 Algebraic Geometry Seminar, Johns Hopkins, Baltimore, March
 2018 Algebraic Geometry Seminar, Univ. of Illinois, Urbana Champaign, November
 Algebraic geometry Seminar, Univ. of Chicago, May

ORGANISATION AND PROFESSIONAL SERVICES

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| 2025 – present | Founding member of the steering committee for European Graduate Research Opportunities for Women (GROW) (yearly conference aimed at female bachelor students for promoting math academia as a career) . |
| 2025 | Initiated the formation of Women* student chapter at the mathematical institute of Leiden. Led by five female students the group organises academic and team-building events open to all students irrespective of gender. |
| Oct. 2025–present | Study Advisor for masters students, Algebra, geometry and number theory, Leiden |
| 2024–present | Member of the EWM |
| July 2025 | Co-organised workshop: K3 surfaces & friends: Brauer groups and moduli, Lorentz Center, Leiden |
| Nov. 2025–present | Masters' study advisor for algebra group, Mathematisch Instituut, Leiden |
| Mar. 2024–present | Consultation (Overleg) team, Mathematisch Instituut, Leiden |

Summer 2024	Co-organised Intercity Geometry Seminar, Amsterdam, Leiden, Utrecht, Antwerp (Informal intercity learning seminar)
Sep. 5–6, 2024	Co-organised: Hodge Theory in Leiden, a conference in honour of Chris Peters' 75th birthday
Dec. 6, 2023	Organised mini-symposium Around Hyperkählers and K3s, Leiden
2022, 2023, 2024	Co-organised the Graduate Research Opportunities for Women (GROW@Bonn) (Conference aimed at female bachelor students for promoting math academia as a career)
July 2021	Served as tutor for Algebraic Geometry Summer School - Hyperkähler and Prym varieties, ICTP
Sep. 2020	Served as a recruitment committee member for W3 professorship in mathematics, Bonn
Until July 2024	Reviewer MathSciNet, zbMATH
Ongoing	Referee and quick opinion Inventiones Mathematicae, Forum of Math (Sigma), PRIMS Kyoto
May 25–27 2018	Co-organised MAGGC (Midwest Algebraic Geometry Graduate Conference), funded partially by the NSF, University of Illinois at Chicago
Fall 2017–Spring 2018	Co-organised Algebraic Geometry Student Seminar, Northwestern University
Nov. 7, 2017	Served as a panel member for TA seminar for 2nd year graduate students, Northwestern University
Winter 2017	Served as a panel member for graduate school applications for NUAWM (Northwestern University Association for Women in Mathematics), an undergraduate club, Northwestern University

MISCELLANEOUS INFORMATION

Computer languages	Haskell, C, HTML, LaTeX
Languages	Fluent in English, Bengali (native), Hindi. General professional proficiency (Goethe-Zertifikat B2) in German. A2 in Dutch.